# Median — Boxplot.

Q: Product sold by 10 Representive:

15, 20, 35, 40, 55, 60, 70, 85, 90.

Now find the Quartiles, $Q_1$, $Q_2$ and $Q_2$ of data.

15, 20, 35, 40, 45, 55, 60, 70, 85, 90 | $\frac{10+1}{2}$

~~Range~~ = ~~Range~~ =

$Q_1 = 35$ | $\frac{n+1}{4}$th =

$Q_2 = 50$ $[(45+55) \div 2]$

$Q_3 = 70$

Ans:

5 Marks

∴ Range = Max − Min

$= 90 - 15 = 75$

Ans:

∴ Inter Quartile range $= Q_3 - Q_1$

$= 70 - 35 = 35$

Ans:

# Median with interval

**Formula**

$$\text{Median} = L + \left(\frac{\frac{n}{2} - pcf}{f}\right) i$$

| Interval | Frequency f | Cumulative Frequency |
|---|---|---|
| 10 - 10 | 5 | 5 |
| 10 - 20 | 12 | 5 + 12 = 17 |
| 20 - 30 | 18 ← f | 17 + 18 = 35 |
| 30 - 40 | 8 | 35 + 8 = 43 |
| 40 - 50 | 3 | 43 + 3 = 46 |

$\sum f = 46$

→ pcf → Preceding cumulative frequency.

→ ~~pcf~~

$$\text{Median} = L + \left(\frac{\frac{n}{2} - cf}{f}\right) i$$

$$= 20 + \left(\frac{\frac{46}{2} - 17}{18}\right) \times 10$$

$$= 20 + 0.333 \times 10$$

$$= 23.33 \text{ (Approx.)}$$

Ans:

**Q:** If asks for Quantile

$$Q_1 = \frac{1}{4} \times 46 = 11.5$$

$$Q_2 = \frac{1}{2} \times 46 = 23$$

$$Q_3 = \frac{3}{4} \times 46 = 34.5$$

Ans:

**Quantile**

$$Q_1 = \frac{1}{4} \times N$$

$$Q_2 = \frac{2}{4} \times N$$

$$Q_3 = \frac{3}{4} \times N$$

Median with interval

## Percentiles:

* Find the 70th percentiles of these data

$$P_{70} = \frac{70}{100} \times 46$$
$$= 32.2$$

Ans:

Percentiles
$P = \frac{k}{100} \times N$

* Find the percentiles of 10th and 50th position

$$P_{10} = \frac{10}{100} \times 46 = 4.6$$

$$P_{50} = \frac{50}{100} \times 46 = 23$$

Ans:

## Deciles:

* Find the deciles of 3rd and 7th positions.

$$D_3 = \frac{3}{10} \times 46 = 13.8$$

$$D_7 = \frac{7}{10} \times 46 = 32.2$$

Ans:

Deciles:
~~Quantile~~

$D_k = \frac{k}{10} \times N$

10 Marks

# Mode — Raw

* Finding mode without interval and frequency with only raw data. ~~like~~ (unorganized)

# Find the Mode from following data

3, 7, 8, 7, 12, 14, 8, 7, 18, 19, 8, 20, 8, 7.

∴ Highes frequent Number = 7 and 8

Mode = 7.8

5 Marks

# Find the mode from 10, 20, 20, 30, 40, 40, 40, 50, 50

Highest frequent number = 40

The Mode is = 40

# Group

| Sale | ~~0-10~~ 5 | 8 $f_1$ | 12 f | 9 $f_2$ | 6 |
|---|---|---|---|---|---|
| Frequency f | 0 - 10 | 10 - 20 | 20 - 30 | 30 - 40 | 40 - 50 |

$$\text{Mode} = L + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2}\right) \times i$$

$$= 20 + \left(\frac{4}{4+3}\right) \times 10$$

$$= 20 + 0.571 \times 10$$

$$= 25.71 \text{ (Approxmatly)}$$

$\Delta_1 = f - f_1 = 4$

$\Delta_2 = f - f_2 = 3$

$L = 20$

$i = 10$

# Standard deviation ~~Mode~~ — ~~group data~~

## Method – 1

* Find the ~~Mea~~ standard deviation from the data, 22, 24, 26, 28, 30,

| $x$ | $\bar{x}$ (mean) | $(x-\bar{x})^2$ |
|---|---|---|
| 22 | 26 | 16 |
| 24 | 26 | 4 |
| 26 | 26 | 0 |
| 28 | 26 | 4 |
| 30 | 26 | 16 |

$\Sigma x = 130$ $\qquad \Sigma(x-\bar{x})^2 = 40$

$$\text{Standard deviation} = \sqrt{\frac{\Sigma(x-\bar{x})^2}{N}}$$

$$= \sqrt{\frac{40}{5}}$$

$$= 2.82842712 \text{ (Approx)}$$

Ans:

N.B = This method will works if there'm no interval and name of this method is Population S.D.

# Standard Deviation + Mean

## Method - 2

Find standard deviation after mean

| Interval | Frequency | Mid x | Deviation d | fd | $fd^2$ |
|---|---|---|---|---|---|
| 0 - 10 | 5 | 5 | -2 | -10 | 20 |
| 10 - 20 | 8 | 15 | -1 | -8 | ~~16~~ 8 |
| 20 - 30 | 12 | 25 ← A | 0 | 0 | 0 |
| 30 - 40 | 7 | 35 | 1 | 7 | 14 |
| 40 - 50 | 3 | 45 | 2 | 6 | 12 |
| | $\Sigma f = 35$ | | | $\Sigma fd = -5$ | $\Sigma fd^2 = 42$ |

$$\text{Mean} = A + \left(\frac{\Sigma fd}{\Sigma f}\right) i$$

$$= 25 + \left(\frac{-5}{35}\right) \times 10$$

$$= 25 + (-0.14) \times 10$$

$$= 23.58 \text{ (Approximately)}$$

$$\text{Standard Deviation} = \sqrt{\frac{\Sigma fd^2}{N} - \left(\frac{\Sigma fd}{N}\right)^2} \times i$$

$$= \sqrt{\frac{42}{35} - \left(\frac{-5}{35}\right)^2} \times 10$$

$$= \cancel{\sqrt{1.78 - 0.29}} \sqrt{1.34 - 0.0204} \times 10$$

$$= 11.38 \text{ (Approx)}$$

# Hermonic Mean

# Define the heromonic mean from data 2, 4, 6, 8, 10.

| $x$ | $\frac{1}{x}$ |
|---|---|
| 2 | 0.5 |
| 4 | 0.25 |
| 6 | 0.1667 |
| 8 | 0.125 |
| 10 | 0.1 |
| | $\Sigma \frac{1}{x} = 1.417$ |

N = Number of Data

$$\text{Hermonic Mean} = \frac{N}{\Sigma \frac{1}{x}}$$

$$= \frac{5}{1.1917}$$

$$= 4.38 \text{ (Approxmatly)}$$

Ans.

# Calculate the Hermonic mean from following data

| Class Interval | 10 - 20 | 20 - 30 | 30 - 40 | 40 - 50 | 50 - 60 |
|---|---|---|---|---|---|
| Frequency (f) | 5 | 10 | 15 | 8 | 2 |

Answer:

| Interval | Frequency | Mid $x$ | $\frac{1}{x}$ | $f \cdot \frac{1}{x}$ |
|---|---|---|---|---|
| 10 - 20 | 5 | 15 | 0.0667 | 0.333 |
| 20 - 30 | 10 | 25 | 0.04 | 0.4 |
| 30 - 40 | 15 | 35 | 0.0286 | 0.429 |
| 40 - 50 | 8 | 45 | 0.0222 | 0.1776 |
| 50 - 60 | 2 | 55 | 0.0182 | 0.0364 |
| | $\Sigma f = 40$ | | | $\Sigma f \cdot \frac{1}{x} = 1.376$ |

$$\text{Hermonic Mean} = \frac{\Sigma f}{\Sigma f \cdot \frac{1}{x}} = \frac{40}{1.376} \approx 29.07$$

Ans.

N.B If the data contain 0 or lower than 0 than not possible to define in Individual data formate math. The math result will appear infinity or Error.

# Boxplot → Range = Max. Num − Lower Num.
→ Inter quartile Range $= Q_3 - Q_1$

# Quartile from table/group data

$Q_1 = \frac{1}{4} \times N$ $\quad Q_2 = \frac{2}{4} \times N$ $\quad Q_3 = \frac{3}{4} \times N$

# Percentiles, $P_k = \frac{k}{100} \times N$

# Deciltiles, $D_k = \frac{k}{10} \times N$

# Median $= L + \left(\frac{\frac{N}{2} - PCF}{f}\right) \times i$

# In raw data mode will be the most frequent value.

# Group data, Mode $= L + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2}\right) \times i$

# Mean $= A + \left(\frac{\Sigma fd}{\Sigma f}\right) \times i$

# Individual Data, Standard Deviation, $S.D = \sqrt{\frac{\Sigma (x - \bar{x})^2}{N}}$

# Group data, $S.D = \sqrt{\frac{\Sigma fd^2}{N} - \left(\frac{\Sigma fd}{N}\right)^2} \times i$

# Hermonic Mean for Individuals, $H.M = \frac{N}{\Sigma \frac{1}{x}}$

# Hermonic Mean for group data, $H.M = \frac{N}{\Sigma f \cdot \frac{1}{x}}$ or $\frac{N}{\Sigma \frac{f}{x}}$